

User Manual

The ASL-DVS Suite

How to use the ASL-DVS web tools day to day: what every page, tab and button does, with a technical close to each chapter for when you want to know how it works. Written for the Pi Zero 2 W build; full-build differences are marked where they matter.

[DASHBOARD](#)[SYSMON](#)[INSTMON](#)[WIFIMON](#)[44HELPER](#)

FILES COVERED BY THIS EDITION

TOOL	FILE	VERSION
Dashboard	<code>asl_dvs_dashboard_pi02w_v9_3_71_13_20261007.py</code>	9.3.71.13-pi02w
SysMon	<code>sysmon_pi02w_v6_13_67_9_20261007.py</code>	6.13.67.9-pi02w
44helper	<code>asl_dvs_m17_44helper_v0_0_175_20261008.py</code>	0.0.175
instmon	<code>instmon_v2_38_3_20261007.py</code>	2.38.3
wifimon	<code>wifimon_v5_28_20261007.py</code>	5.28
Installer	<code>install_asl_dvs_v6_6_20261006.sh</code>	6.6
SysMon FULL	<code>sysmon_v6_13_74_20261007.py</code>	6.13.74
Dashboard FULL	<code>asl_dvs_dashboard_v9_3_73_20261006.py</code>	9.3.73

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How to use this manual

If you are new to the suite, read Part I first, then turn to the chapter for the tool in front of you. Every chapter after Chapter 2 follows the same order, so you always know where to look:

- 1 What it's for:** what the tool or tab does, and when you would open it.
- 2 Before you start:** what has to be installed, and anything that can go wrong.
- 3 Tab by tab:** each card and button, and what good and bad readings look like.
- 4 Common tasks:** short recipes, step by step.
- 5 Under the hood:** the technical close, in a purple-labelled panel. It covers the files, services, ports and timers involved. You can skip it on a first read.

Markers used in the text

NOTE Background, or something worth knowing.

TIP A quicker or safer way to do something.

CAREFUL This changes your node, restarts something or needs a reboot.

FULL BUILD Applies to the full build (Pi 3, 4 or 5) only. Also shown as the badge **FULL**.

Commands and pictures

Commands are typed on the Pi, in an SSH session or at a keyboard and screen. Lines starting with **#** are notes. Replace `<pi-address>` with your Pi's IP address or host name.

```
# a note about the next line
sudo systemctl status sysmon
```

Pictures marked **Illustration** are drawn to show a page's layout. They use the same tab names and order as the real page, but the values in them are made up. Example identities use the placeholders `NØCALL` and `<NODE>`.

PART I

Getting started

What the suite is, what each tool is for, how to log in, and the short version of installing and updating it. The full install steps are in the install manuals in the [docs/](#) folder.

1. The suite at a glance
2. Installing and updating, in brief

The suite at a glance

ASL-DVS is a set of five small web tools that run on the same Raspberry Pi as your AllStarLink node. Each tool has its own page in the browser. Together they let you run, check, repair and update the node without typing commands.

1.1 What is on the node

The tools sit on top of a working AllStarLink 3 (ASL3) node with DVSwitch. It helps to know the main parts, because the tools check and talk to all of them:

PART	WHAT IT DOES
Asterisk + app_rpt	The ASL3 engine. It runs your node, links to other AllStarLink nodes over IAX2 and handles EchoLink.
Analog_Bridge	Turns the node's audio into digital voice and back again. It talks to Asterisk on a private node (1999 by convention).
MMDVM_Bridge	Connects Analog_Bridge to the digital networks: DMR directly, and System Fusion, P25, NXDN and D-STAR through their gateways.
Gateways	YSFGateway, P25Gateway, NXDNGateway and ircDDBGateway. Each one connects to that mode's reflectors or talkgroups.
USRP2M17	Bridges a second private node (1917 by convention) to an M17 reflector.

```
AllStarLink network
| IAX2 (UDP 4569)
[ ASL3 / Asterisk ]----- your node <NODE>      (radio, optional)
| USRP (UDP, this Pi only)
+-- private node 1999 -- Analog_Bridge -- MMDVM_Bridge -- DMR network
|                                     |
|                                     YSFGateway / P25Gateway / NXDNGateway / ircDDBGateway
|
+-- private node 1917 -- USRP2M17 ----- M17 reflector
```

Figure 1.1 The audio path through a bridge node. Node numbers 1000 to 1999 are private: they never leave the Pi. The full layout is in [docs/ASL-DVS_Idealized_Node_Configuration.pdf](#).

1.2 The five tools

Each tool is one file, runs as its own service, and serves its own page on its own port.

TOOL	PORT	WHAT IT'S FOR	OPEN IT WHEN YOU WANT TO...
Dashboard	8989	Node control.	Connect to AllStar or EchoLink nodes, change DMR talkgroup or YSF, P25, NXDN, D-STAR, XLX or M17 reflector, use favorites, make a phone call.
SysMon	9999	System monitor and health checks.	See whether the Pi and node are healthy, find a stopped service or a bad config file, read logs, look after the SD card.
44helper	9997	Setup, repair and configuration helper.	Set up 44Net or Cloudflare remote access, install ASL3, DVSwitch, SVXLink, M17 or Ampersand, run updates, save and restore your config.

instmon	8990	Installer and manager for the suite.	Update the tools from GitHub, start or stop them, pause the suite for a big job, back up or image the SD card.
wifimon	8991	WiFi and power watchdog.	Change WiFi networks, check signal and supply voltage, set up the automatic safe shutdown.

NOTE 44helper is optional. The other four make up the core suite. A sixth item, the Watchdog script, is no longer needed: the Dashboard's own service restarts it if it stops.

Which tool for which job

I WANT TO...	GO TO
Talk on a DMR talkgroup or a reflector	Dashboard, the mode's tab
Link to another AllStar node	Dashboard, ASL tab
Find out why something stopped working	SysMon, Overview and Services, then the matching check tab
Check that my config files agree with each other	SysMon, ASL-DVS and DVSM tabs; 44helper, Config tab
Save my config before I change it	44helper, Config tab (restore point)
Reach my node from outside my home network	44helper, Router Install / Tunnel tabs or Cloudflare
Update the suite	instmon, GitHub Updates
Update Linux and ASL packages	44helper, Update tab
Make the SD card last longer	SysMon, SD Card tab; 44helper, System Optimization
Change WiFi network	wifimon

1.3 A first look at each page

Every page has a header with the tool's name and version and a Log Out button. Most pages have a row of tabs under the header; click a tab to switch view. The pictures below show each tool's tabs in order.

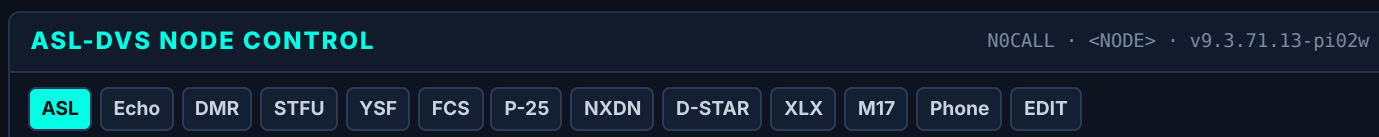


Figure 1.2 · Illustration. The Dashboard (port 8989). One tab per mode, plus Phone and EDIT. EDIT holds your favorites and settings. The Pi Zero 2 W build keeps every tab of the full build.

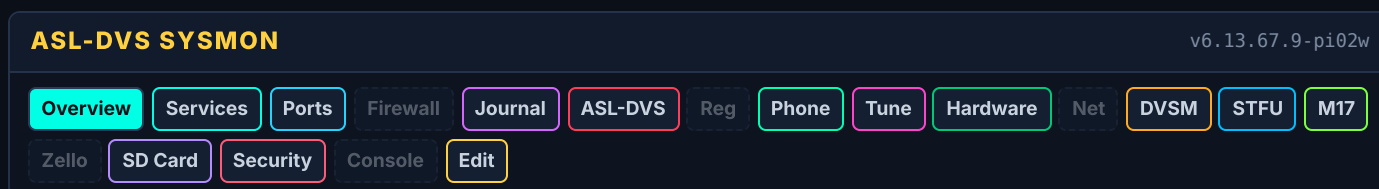


Figure 1.3 · Illustration. SysMon (port 9999). The 14 solid tabs are in the Pi Zero 2 W build. The 5 faded tabs (Firewall, Reg, Net, Zello, Console) are **FULL** build only.

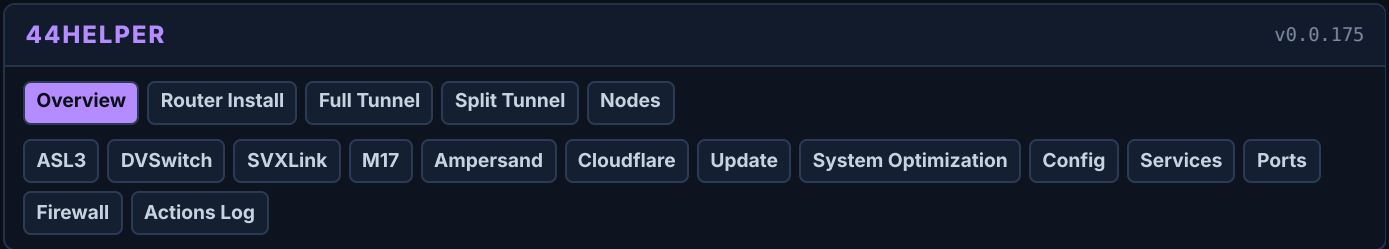


Figure 1.4 · Illustration. 44helper (port 9997). The first row is 44Net remote access. The second row has the installers, maintenance, Config and the status tabs. 44helper is the same on every Pi.

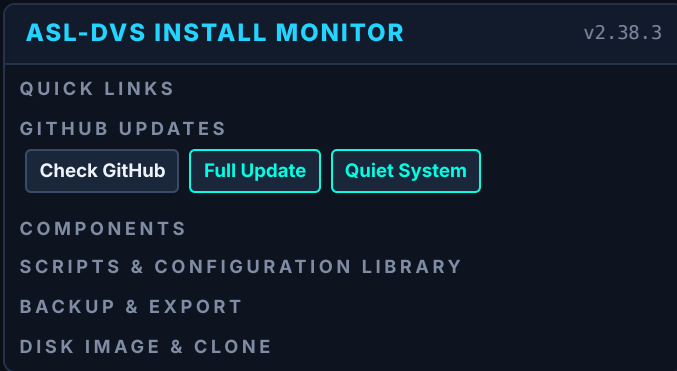


Figure 1.5 · Illustration. instmon (port 8990) is one long page of sections, not tabs.

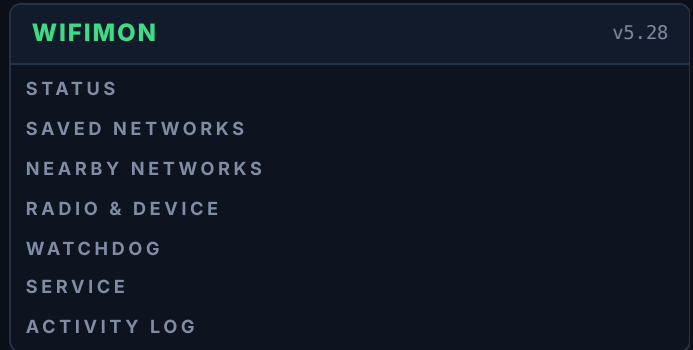


Figure 1.6 · Illustration. wifimon (port 8991) is also one page of cards.

1.4 Two builds: Pi Zero 2 W and full

The Dashboard and SysMon come in two builds. The other tools have a single build that runs on every Pi.

PI ZERO 2 W BUILD PI02W For the 512 MB Pi Zero 2 W. This manual is written for it. • Dashboard: every tab is kept. It searches the AllStar and EchoLink lists from disk instead of memory, and refreshes less often. • SysMon: drops the Firewall, Reg, Net, Zello and Console tabs to save memory.	FULL BUILD FULL For a Pi 3, 4 or 5. • Dashboard: the same tabs, and it keeps the lists in memory. • SysMon: all 19 tabs. Chapter 6 covers the five extra ones.
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Which build am I running? Look at the version in the page header. A Pi Zero 2 W build version ends in -pi02w, for example 6.13.67.9-pi02w. A full build version doesn't, for example 6.13.74.

NOTE Apart from SysMon, the builds look and work the same, so this manual points out the full build only where it differs.

1.5 Logging in

- 1 Open the page in a browser: http://<pi-address>:<port>, for example http://192.168.1.50:9999 for SysMon.
- 2 Type the Pi's **root password** and press Log In. There is no user name.
- 3 When you're done, press **Log Out** in the header, especially on a shared computer.

- **The Dashboard and SysMon share one login.** Log in to either one and the other opens without asking, in the same browser. A shared login lasts 12 hours from the last time you used either page.
- **instmon, wifimon and 44helper each have their own login.**
- **Wrong passwords lock you out for a while.** On the Dashboard and SysMon, five wrong passwords within 5 minutes lock that address out of both for 15 minutes. The other tools have similar limits. Wait it out; don't keep guessing.
- **44helper stays logged in** until you log out, press **Log out all devices**, or the Pi reboots, because its jobs can run for a long time.

CAREFUL The pages use plain HTTP, so your password crosses the network unencrypted. Only open them on your own network or over a VPN or tunnel, never straight from the internet. Chapters 7 and 8 cover safe remote access with 44helper.

TIP If the login is always refused, set the root password again on the Pi with `sudo passwd root`. Raspberry Pi OS ships with root locked, so a new Pi needs this once.

1.6 Phones and tablets

The pages are laid out for phone, tablet and desktop widths. On a phone, wide tables hide their less important columns, and the tab row wraps onto more lines. Turning a tablet to landscape shows everything.

Leave a page open only while you're using it. Each open page polls its tool for fresh data, and on a Pi Zero 2 W that costs CPU and memory.

UNDER THE HOOD

Services, files and ports

TOOL	SYSTEMD SERVICE	PROGRAM	CONFIG
Dashboard	asl_dvs_dashboard	/usr/local/bin/asl_dvs_dashboard.py	/etc/asl_dvs/asl_dvs.conf
SysMon	sysmon	/usr/local/bin/sysmon.py	/etc/sysmon/sysmon.conf
44helper	44helper	/opt/44helper/asl_dvs_m17_44helper.py	/etc/44helper/44helper.conf
instmon	instmon	/usr/local/bin/instmon.py	/etc/asl_dvs/instmon_library/
wifimon	wifimon	/usr/local/bin/wifimon.py	/etc/wifimon/wifimon.conf

The Dashboard and SysMon program paths are links into `/usr/local/lib/asl_dvs/` and `/usr/local/lib/asl_dvs_sysmon/`. The installed file keeps its own name, so the version is always visible on disk.

One process each, started through the launcher

Every service starts its tool through the shared launcher `/usr/local/bin/asl_dvs_launch.py`:

```
ExecStart=/usr/bin/python3 /usr/local/bin/asl_dvs_launch.py /usr/local/bin/sysmon.py
```

The launcher loads the tool as a module so that Python keeps its compiled copy in `__pycache__`. That roughly halves each tool's memory (SysMon on a Pi Zero 2 W: 54 MB started directly, 26 MB through the launcher). The launcher also turns on three memory savers: `python3 -O0`, `MALLOC_ARENA_MAX=2` and a periodic `malloc_trim`. Each has an off-switch file in `/etc/asl_dvs/` named `launch_no_optimize`, `launch_no_arena_cap` or `launch_no_trim`.

How a page works

- Each tool is a small Python HTTP server. The page's HTML, CSS and JavaScript are built into the tool's file, so there is nothing else to install and no web server to configure.
- The page fetches live data from the tool's own `/api/...` addresses as JSON and redraws its cards.
- The login checks the password against the root entry in `/etc/shadow`, and sets a session cookie. Requests that change anything also need a matching CSRF header, so another web page can't act for you.
- Version numbers come from the `VERSION` line in each file. The `-pi02w` suffix is how the installer and instmon recognise the Pi Zero 2 W build.

Checking on a tool from the shell

```
# is it running, and its last 30 log lines
systemctl status sysmon --no-pager
journalctl -u sysmon -n 30 --no-pager
# is anything listening on its port?
ss -tlnp | grep 9999
# restart all the web tools
sudo systemctl restart sysmon asl_dvs_dashboard instmon wifimon 44helper
```

Installing and updating, in brief

This chapter covers just enough to get the suite onto a node and keep it current. For every step and option, use the install manual for your build.

MANUAL	FILE IN DOCS/
Pi Zero 2 W install	ASL-DVS_Pi02w_Install_Manual.pdf
Full build install FULL	ASL-DVS_Full_Build_Install_Manual.pdf

2.1 Before you start

- A Raspberry Pi already running AllStarLink 3 with DVSwitch set up.
- SSH access with `sudo`, and a root password (`sudo passwd root`).
- On a Pi Zero 2 W, some free memory. Check with `free -m`. The install manual lists services you can turn off if you don't use them.

2.2 Install in three steps

- 1 Get the files.** Download the current files from github.com/kd8pgk/ASL-DVS into one folder on the Pi.
- 2 Install instmon first.** It can then install and update everything else from the browser.

```
sudo python3 instmon_v2_38_3_20261007.py --install
```

- 3 Install the rest** with the installer. On a new node it asks for your callsign, node number and a label.

```
sudo bash install_asl_dvs_v6_6_20261006.sh
```

The installer picks the build for you: the Pi Zero 2 W build on a Pi Zero 2 W, the full build on anything else. If one build is already installed, it keeps that build. When it finishes, it checks that each tool is running and listening on its port.

TIP 44helper is optional. Install it the same way, or from instmon's page, when you want its remote-access, installer or Config tools.

2.3 First visit to each page

PAGE	ADDRESS	CHECK FIRST
Dashboard	<a href="http://<pi-address>:8989">http://<pi-address>:8989	Your callsign and node number are right in the header.
SysMon	<a href="http://<pi-address>:9999">http://<pi-address>:9999	The Overview shows no red items (Chapter 3).
instmon	<a href="http://<pi-address>:8990">http://<pi-address>:8990	Every tool shows as installed and running.
wifimon	<a href="http://<pi-address>:8991">http://<pi-address>:8991	Answer "Turn the watchdogs on?" (below).
44helper	<a href="http://<pi-address>:9997">http://<pi-address>:9997	Take a restore point on the Config tab (Chapter 11).

wifimon's watchdogs

- A new wifimon starts with both watchdogs off: it shows WiFi and voltage, but won't act on them. On your first login it asks whether to turn them on. **Add your WiFi networks first.**
- **WiFi watchdog:** reconnects to your saved networks, and shuts the Pi down after 3 minutes with no connection.
 - **Voltage watchdog:** shuts the Pi down after 30 seconds of low supply voltage.

CAREFUL Both watchdogs can shut the node down. A Pi Zero 2 W is sensitive to voltage drop, so use a good power supply and a short cable before turning the voltage watchdog on.

2.4 Keeping the suite up to date

Updates come from GitHub through instmon's **GitHub Updates** section.

GITHUB UPDATES

Check GitHub

Full Update

3 newer on GitHub · checked 14:02

Pi Zero 2 W build:

Pi02w fork (default)

Full build (override)

TOOL	INSTALLED	GITHUB	
SysMon	6.13.67.8-pi02w	6.13.67.9-pi02w	NEWER
44helper	0.0.174	0.0.175	NEWER
SysMon	–	6.13.74	other build

Quiet System

Restore

Figure 2.1 • Illustration. instmon's GitHub Updates section on a Pi Zero 2 W. The build choice appears only on a Pi Zero 2 W.

BUTTON	WHAT IT DOES
Check GitHub	Lists the newest copy of each tool on GitHub, and marks the ones newer than yours.
Update	Downloads and checks one file into instmon's library. Nothing is installed until you press Install on it.
Full Update	Updates every installed tool in one go, one tool at a time, with automatic roll-back if a new version fails.
Quiet System	Pauses SysMon, the Dashboard and 44helper during a big upload or install. The radio side keeps running, and everything comes back by itself after 30 minutes. Restore brings it back sooner.

TIP Before a Full Update, take a restore point in 44helper's Config tab. Full Update only replaces the tools, but a restore point means you can roll back any config change you make while trying out the new versions.

Updating Linux, ASL3 and DVSwitch packages is a separate job. 44helper's Update tab handles it (Chapter 10).

2.5 Removing the suite

```
# remove one tool
sudo python3 /usr/local/bin/sysmon.py --uninstall
# remove every tool except instmon
sudo bash uninstall_asl_dvs_all_v1_3.sh
```

Your settings in `/etc/asl_dvs/`, instmon's library and wifimon's config stay in place. Remove instmon last, with its own `--uninstall`.

What install and update actually do

--install, per tool

- Copies the file into place, writes its `.service` unit, and writes the shared launcher if it is missing.
- Enables the service so it starts at boot, and starts it.
- `--uninstall` stops and removes the service and the program. It removes the launcher only once no other service uses it.

How the installer picks a build

1. `SYSMON_VARIANT` / `DASH_VARIANT` set to `pi02w` or `full`, if you set them;
2. otherwise, the build already installed;
3. otherwise, `pi02w` on a Pi Zero 2 W and `full` on any other Pi.

If no file of the chosen build is found, that tool is skipped with a warning; the other build is never used in its place. The installer looks in its own folder, the current folder, `/root`, `/home/pi`, your home folder, `/tmp` and instmon's library. `--non-interactive` (or `-y`) skips every question and takes the identity from `AUTO_CALLSIGN`, `AUTO_NODE` and `AUTO_LABEL`.

The checks on every download

Before instmon saves a file from GitHub, the file must match GitHub's file list. It must also have a version line and the right first line, pass a syntax check, and support `--install`.

Full Update, step by step

1. Downloads and checks every new file before anything changes.
2. Pauses the web tools.
3. For each tool in turn, saves a copy of the installed file in the library, runs the old `--uninstall`, and runs the new `--install`. If the new one fails, it puts the old one back and moves on.
4. Updates instmon last. A timer reinstalls the old instmon if the new one isn't answering after 90 seconds.
5. Ends with a summary. New install scripts go only to the Scripts library and are never run for you.

On a Pi Zero 2 W, the build choice decides which build Full Update installs. A tool installed in the other build is listed as **SWITCH** and replaced. On other Pis the choice is hidden and each tool keeps its build.

Settings that survive an update

Config files in `/etc` are never touched by an update. SysMon's Hardware tab options (HDMI and video driver off, lower GPU memory) are saved when SysMon is uninstalled during an update and put back on install, if the install follows within 30 minutes.

PART II

SysMon

SysMon is the node's health monitor. It shows whether every service is up, checks your Asterisk, DVSwitch and M17 config files against each other, reads logs, and looks after the hardware and SD card. Most of what it does is read-only. Every button that changes something says so.

- 3. SysMon basics and live status
- 4. Node and DVSwitch health
- 5. System care
- 6. Full-build tabs

SysMon basics and live status

This chapter covers the parts of SysMon you see every time: the header and command bar, then the four status tabs: Overview, Services, Ports and Journal. Open SysMon at <http://<pi-address>:9999>.

3.1 What SysMon is for

Open SysMon when something on the node isn't working, or when you want to be sure everything is. In one place it shows:

- whether Asterisk, the DVSwitch bridges, the gateways, USRP2M17 and the ASL-DVS tools are running;
- the Pi's load, memory, temperature and power health, updated every 5 seconds;
- which programs are listening on which network ports, and whether two of them clash;
- the system log (journal) of any service, filtered and ready to copy.

The status tabs can also start, stop and restart services. Everything else in SysMon is either read-only or asks before it changes a file.

3.2 The page at a glance

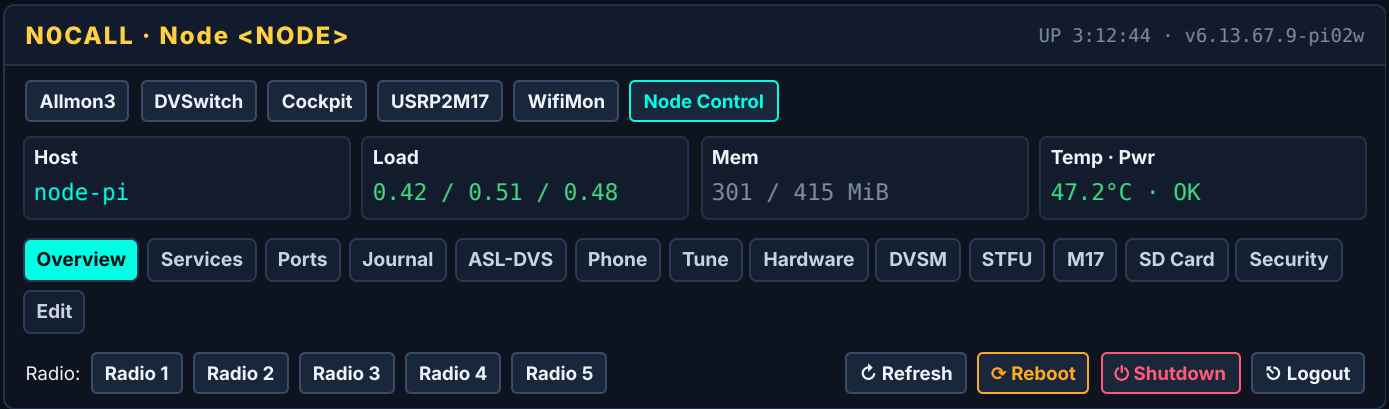


Figure 3.1 · Illustration. SysMon's top section. From the top: header, quick links, status strip, tabs and command bar.

PART	WHAT IT SHOWS OR DOES
Header	Your callsign and node number, how long the Pi has been up, and SysMon's version.
Quick links	Open Allmon3, the DVSwitch page, Cockpit, the USRP2M17 page and wifimon in a new tab. Node Control goes back to the Dashboard.
Status strip	Host name, load, memory, CPU temperature and power health (below).
Tabs	One per view. You can hide the ones you don't use on the Edit tab (Chapter 5).
Radio 1 to 5	Saved sets of radio audio settings. Pressing one applies it (Chapter 5, Tune tab).
Refresh	Reads the status strip again now, instead of waiting for the next 5-second update.
Reboot / Shutdown	Restart or power off the whole Pi. Both ask first. After Shutdown, the Pi stays off until you remove and restore power.

Reading the status strip

FIELD	GREEN	AMBER	RED
Load (1, 5, 15 min)	1-minute load up to 1.5	above 1.5	above 3
Temp	up to 60 °C	above 60 °C	above 75 °C
Pwr	OK	WARN : a problem happened since boot but isn't happening now	⚡ UV under-voltage now, or ⚡ THR throttled now

Hover over **Pwr** to see exactly what was flagged, such as "Under-voltage since boot". When under-voltage starts, SysMon also shows a pop-up warning to check the power supply.

TIP A Pi Zero 2 W has four 1 GHz cores and 512 MB, so a load of 1 to 2 during a tune or an update is normal. A load that *stays* red with nothing going on usually means a service is stuck restarting. Check the Overview for an NR: count.

The status dots

Every service list in SysMon uses the same coloured dots:

DOT	MEANS
● Active	Running normally.
● Stopped	Installed but not running. That is fine for a gateway you don't use, and a problem for Asterisk.
● Failed / Masked	It crashed or failed to start, or it has been blocked from starting (masked).
● Unknown	SysMon couldn't read its state.
● Not installed	There is no such service on this Pi. That is normal for optional parts.

Two banners you may see

- **"Restart sysmon to apply server address changes"**: you changed SysMon's port or bind address on the Edit tab. It takes effect when SysMon restarts.
- **"wifimon — Node Shutting Down"**: wifimon has started a safe shutdown because of low voltage or lost network. The Pi needs power-cycling, or its WiFi back, before it will come up again.

3.3 Overview tab

The Overview is SysMon's home page. It lists your **pinned services** in groups. Out of the box the groups are System, ASL-DVS, Asterisk Core and M17, which cover everything a bridge node runs.

ASTERISK CORE	OWNER	PERMS	STATUS	PORT					
● asterisk	asterisk	644	active	:4569/udp	[OK]	↺ Restart	■ Stop	💀 Kill	Edit
● analog_bridge	root	644	active			↺ Restart	■ Stop	💀 Kill	Edit
● ysfgateway	mmdvm	644	failed	NR:5▲		▶ Start	× Reset	Edit	
● p25gateway	mmdvm	644	inactive			▶ Start	Edit		
● quantar_bridge	—	—	not-inst						
Global Actions						📄 Copy	■ Stop All	↺ Restart All	

Figure 3.2 • Illustration. One Overview group. The buttons on each row depend on the service's state.

COLUMN	WHAT IT TELLS YOU
Name	The service. Click it to open its detail panel (3.4).
Owner	The Linux user it runs as. Amber means it runs as root but doesn't need to. Click the amber value for the Security tab's explanation.
Perms	The permissions on its service file. Amber or red means other users could change it. Click for the explanation.
Status	<code>active</code> , <code>inactive</code> , <code>failed</code> , <code>not-inst</code> or <code>unknown</code> .
Port	The port it should listen on. If it has restarted on its own since it was last started, this shows <code>NR:</code> and the count instead, with ▲ above 3.
[OK] / [DUP]	For a running service with a port: <code>[DUP]</code> means another program has the same port and protocol, and hovering names both. <code>[OK]</code> means no clash.

Row buttons

STATE	BUTTONS
active	Restart ; Stop (asks first); Kill , which ends the process at once with no clean shutdown (asks first); Edit , which opens the service file.
failed	Start ; Reset , which clears the failed mark and the restart counter; Edit .
inactive	Start ; Edit .

Restart is done carefully: SysMon stops the service, waits for it to stop, starts it, and waits for it to come up. The Overview refreshes itself a few times over the next half minute while that happens.

Global Actions

- **Copy** puts the whole Overview on the clipboard as text, ready to paste into a help request.
- **Stop All** stops every pinned service except SysMon itself.
- **Restart All** restarts every pinned service, SysMon included.

CAREFUL **Stop All** and **Restart All** act at once, with no confirmation. Stop All takes the node off the air: it stops Asterisk, the bridges and the Dashboard. To get it back, press **Restart All** or reboot. SysMon keeps running, so the page stays usable.

3.4 Services tab

The Services tab lists **every** service on the Pi, not just the pinned ones.

- **Active** shows only running services. **All** also shows stopped, failed and disabled ones.
- Type in **filter by name** to narrow the list, for example `gateway`.
- The list refreshes every 30 seconds while the tab is open. **Copy** puts it on the clipboard as text.

The service detail panel

Click any service, here or on the Overview, to open its panel. The top shows badges for its state, whether it starts at boot, its process number, its restart count (`NR:`), CPU and memory use, and the user it runs as. Under that, **Ownership & permissions** shows who owns the service file and the program.

SECTION	BUTTON	WHAT IT DOES, IN PLAIN WORDS
Daemon Controls	Reload Daemon	Makes systemd re-read every service file. Needed after you edit one by hand.
	Reset Failed	Clears a "failed" mark so the service can be started cleanly.
	Mask / Unmask	Mask blocks a service from starting at all, even by hand, until you Unmask it. Asks first.
Runtime Controls	Start / Stop / Restart	As on the Overview. Stop asks first.
	Reload	Asks the program to re-read its settings without restarting. Greyed out for services that don't support it.
Boot Persistence	Enable / Disable	Whether the service starts by itself at boot. Disable asks first. Neither one starts or stops it now.
Pinned Services	Pin / Unpin	Adds the service to the Overview, in a group you pick or a new one, or removes it.
Status	(output)	The service's current status and its last 50 log lines, with errors in red and warnings in amber.

Editing a service file

Edit opens the service's systemd file in an editor, with **Save**, **Restart**, **Reload Daemon** and **Copy** buttons. After saving, press Reload Daemon and then Restart for the change to take effect.

NOTE Service files that came with a package live in `/lib/systemd/system`. SysMon never changes those. Saving one creates your own copy in `/etc/systemd/system/`, which takes priority and survives package updates. A yellow bar warns you when this happens.

3.5 Ports tab

The Ports tab lists every network port a program on the Pi is listening on. It refreshes every 15 seconds.

COLUMN	MEANING
Dot	Green: the port belongs to a known service. Grey: a listener SysMon can't tie to a service.
Port / Proto	The port number, and TCP or UDP.
Process / PID	The program listening, and its process number.
Known Service	The service that program belongs to, or "— unknown —".
Conflict	<code>[OK]</code> , or <code>[DUP]</code> if more than one program has the port.

- **All / TCP / UDP** filter the list. **Copy table** copies what you see; **Copy all** copies every port.
- Click a row for its panel: the listen address, **Probe** buttons (HTTP and TCP for a TCP port, UDP for a UDP port) that test whether it answers, and for a known service **Restart Service**, **Stop Service** and **Journal**.

NOTE [OK] only means nothing else is fighting for that port. It doesn't mean the service is healthy. A DVSwitch port clash is a common cause of "no audio". The pairing rules are in [docs/ASL-DVS_Idealized_Node_Configuration.pdf](#), section 4.

3.6 Journal tab

The Journal tab lists your pinned services that are installed, followed by any other running service. Click one to open its log in a pop-up window.

CONTROL	WHAT IT DOES
Lines 50 / 200 / All	How far back to read. All means the last 2,000 lines.
grep filter	Shows only lines containing your text, for example <code>error</code> or a talkgroup number.
Refresh	Reads the log again.
Restart / Start	Restart or start the service from the log window, so you can watch what happens.
Copy	Copies the lines you see (after the filter).

3.7 Common tasks

A service shows red

- 1 On the Overview, note the `NR:` count. A high count means it keeps crashing.
- 2 Open the Journal tab, click the service, choose 200 lines and filter on `error`.
- 3 Fix the cause. It is usually a config file, covered on the ASL-DVS, DVSM, STFU and M17 tabs (Chapter 4).
- 4 On the Overview, press **Reset**, then **Start**. Watch the dot turn green.

Find out what is using a port

- 1 Open the Ports tab and press UDP or TCP.
- 2 Find the port number. The Process and Known Service columns name the owner.
- 3 If it shows `[DUP]`, hover over it for both programs, then stop or reconfigure one of them.

Add a service to the Overview

- 1 Services tab, press **All**, and filter for the service.
- 2 Click it, pick a group under Pinned Services (or "New group...") and press **Pin**.

Send someone a status report

Use **Copy** on the Overview, the Services list, a Ports table or a journal window, and paste it into your message. Copies are plain text, so they paste cleanly into an email or forum post.

How SysMon gathers and acts

Refresh timings

WHAT	HOW OFTEN	NOTES
Background poll (server)	every 10 s	Host, load, memory, swap, temperature, voltage, throttle state, disk, SD health, process count, wifimon shutdown state. Also tells systemd's watchdog that SysMon is alive.
Quick tests, port clashes	every 30 s	Every third background poll.
Status strip (page)	every 5 s	<code>GET /api/status</code> , read from the background poll's snapshot, so it costs almost nothing.
Services list	every 30 s	Only while the Services tab is open.
Ports list	every 15 s	Only while the Ports tab is open.
M17 reflector list	hourly check	Downloads <code>M17Hosts.json</code> again when it is stale (Chapter 4).

Switching tabs stops the previous tab's timer, so only the tab you're looking at polls.

Where the data comes from

- **Services:** `systemctl show` and `systemctl list-units`. The owner comes from the running process, or from `User=` in the unit.
- **Ports:** `ss -tlnpu`, with each process's command line from `/proc/<pid>/cmdline`.
- **Journal:** `journalctl -u <unit> -n <lines>`, between 10 and 2,000 lines.
- **Power:** `vcgencmd get_throttled`. Its bits give under-voltage, frequency cap, throttling and temperature limit, both now and since boot. Without it, SysMon shows the core voltage instead.

Service actions

Every action goes to `POST /api/svc`. Unit names are checked before anything runs.

- **Simple actions** (start, stop, reset-failed, enable, disable, reload, mask, unmask) run the matching `systemctl` command, with a 15-second limit.
- **Restart** runs in the background: `systemctl stop`, wait up to 15 s; if it is still running, `SIGKILL` and wait up to 5 s; then `systemctl start` and wait up to 10 s for it to become active. The result goes to SysMon's own log.
- **Kill** sends `SIGKILL` to the main process. systemd may then restart it, depending on the unit's `Restart=` setting.
- **Stop All** runs `systemctl stop` on each pinned unit except `sysmon.service`. **Restart All** runs `systemctl restart` on each pinned unit, SysMon included.

The pinned list

The Overview groups come from `[services] pinned` in `/etc/sysmon/sysmon.conf`, one line per entry:

```
__GROUP__:Asterisk Core
asterisk.service | ASL3 Asterisk/app_rpt | /etc/asterisk/rpt.conf | 4569 | udp
# unit | description | config_path | port | proto    ( - for none )
```

You can edit it as text on the Edit tab (Chapter 5). Pin and Unpin rewrite it for you.

Login and process

- SysMon runs as root, because it reads every config file and controls services. It refuses to start as any other user.
- SysMon and the Dashboard keep their logins in one shared file, `/run/asl_dvs/auth_session.json` (mode 600, locked while written). Both use the cookie `asl_dvs_session`, so one login covers both pages. A session expires 12 hours after it was last used. Five failures from one address within 5 minutes lock it out of both for 15 minutes. `/run` is cleared at reboot, so a reboot logs everyone out.
- The password is checked through PAM, with `/etc/shadow` as a fallback.
- The page is built once at start-up and sent compressed. Everything after that is small JSON requests.

```
# SysMon's own checks, without the web page
sudo python3 /usr/local/bin/sysmon.py --selftest
```